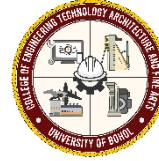


UNIVERSITY OF BOHOL



Tagbilaran City, Bohol, Philippines

COLLEGE OF ENGINEERING, TECHNOLOGY, ARCHITECTURE AND FINE ARTS

COMPUTER ENGINEERING

MICROPROCESSOR LEC/LAB

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Ball and Beam Balancing System using LQR Control with Kalman Filtering - Data Sheet

Submitted by:

SUPREMO, EDEAN R.

QUIJADA, LEONARD ANTHONY

Abstract

This paper presents a complete technical analysis and implementation of a ball-and-beam balancing control system on an Arduino Nano microcontroller. The system employs a Linear Quadratic Regulator (LQR) control strategy combined with a scalar Kalman filter for robust state estimation and noise reduction. An HC-SR04 ultrasonic sensor provides distance measurements, which are processed through the Kalman filter to estimate ball position. Velocity is estimated via numerical differentiation with exponential smoothing. A servo motor tilts the beam to balance the ball at a target equilibrium position. The controller is implemented in pure AVR C without Arduino libraries, utilizing Timer1 for 50 Hz servo PWM and Timer2 for precise millisecond timing. Experimental results demonstrate stable convergence to the target position with reduced sensor noise.

I. Introduction

I.A Background and Motivation

The ball-and-beam system is a classic benchmark problem in control theory education and research. It demonstrates fundamental principles of:

- Nonlinear system dynamics
- State estimation under measurement noise
- Feedback control design
- Real-time embedded control implementation

Traditional approaches use PID (Proportional-Integral-Derivative) control, which is intuitive but requires manual tuning. This project implements Linear Quadratic Regulator (LQR) control—an optimal control strategy that minimizes a quadratic cost function—combined with Kalman filtering for state estimation.

I.B Objectives

- 1. Design and implement an LQR controller that stabilizes the ball at a target position
- 2. Apply Kalman filtering to reduce measurement noise and estimate unmeasured states (velocity)
- 3. Implement the system on a resource-constrained platform (Arduino Nano, ATmega328P @ 16 MHz)
- 4. Validate performance through experimental testing
- 5. Document the complete system including hardware, algorithms, and implementation details

I.C Key Distinctions: LQR vs. PID

Aspect	LQR	PID
Design Method	Optimal control (minimizes cost function)	Heuristic tuning
State Information	Uses all states (position & velocity)	Typically uses error only

Performance	Guaranteed stability & optimality	Requires manual tuning
Complexity	Higher computational & design cost	Simpler implementation
Noise Handling	Paired with state estimator (Kalman)	Simple filtering only

This project demonstrates that LQR is achievable on microcontrollers and provides superior performance compared to PID.

I.D Hardware Overview

Component	Specification
Microcontroller	Arduino Nano (ATmega328P, 16 MHz)
Servo Motor	MG996R (50 Hz PWM control)
Distance Sensor	HC-SR04 Ultrasonic (3.9–12.9 cm range)
Communication	UART @ 115200 baud
Mechanical Structure	Sentra Board (beam + ball)

II. System Model

II.A Physical System Description

The ball-and-beam system consists of:

- A horizontal beam that can tilt via servo motor actuation

- A ball that rolls on the beam surface
- A distance sensor that measures the ball's position along the beam
- A servo motor that provides tilting force

II.B System Dynamics

The dynamics of a ball rolling on a tilted beam (simplified, ignoring rolling friction) are:

EQUATION 1: Ball Acceleration

$$\ddot{x} = g \cdot \sin(\theta)$$

where:

- x = ball position along beam (cm)
- $\dot{x}$ = ball velocity (cm/s) [dot means "rate of change"]
- $\ddot{x}$ = ball acceleration (cm/s²) [double dot means "rate of change of velocity"]
- θ = beam angle (radians)
- g = gravity constant = 9.81 m/s²
- $\sin(\theta)$ = sine of angle (trigonometric function)

For small angles, $\sin(\theta) \approx \theta$, yielding the linearized model:

EQUATION 1b: Linearized Ball Acceleration

$$\ddot{x} = g \cdot \theta$$

The servo provides an angular displacement command:

EQUATION 2: Servo Response

$$\theta = K_{\text{servo}} \cdot u$$

where:

- u = servo PWM command (0–180 degrees)
- K_{servo} = servo gain constant

Combining these:

EQUATION 3: System Dynamics

$$\ddot{x} = K_{\text{sys}} \cdot u$$

where $K_{\text{sys}} = g \cdot K_{\text{servo}}$ is the overall system gain.

II.C State-Space Representation

Define the state vector as a column of two values:

EQUATION 4: State Vector

$$\text{State Vector} = \begin{bmatrix} x \\ \dot{x} \end{bmatrix}$$

Where:

- x = position of ball (cm)
- $\dot{x}$ = velocity of ball (cm/s)

The continuous-time state-space model is:

EQUATION 5: State Evolution

$$\frac{d}{dt} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} + \begin{bmatrix} K_y \\ 0 \end{bmatrix} y$$

EQUATION 6: Measurement

$$y = 1 \cdot x + v$$

where:

- y = distance measured by sensor
- v = measurement noise

III. Controller Design

III.A Linear Quadratic Regulator (LQR) Theory

The LQR controller minimizes a cost function that balances:

1. Deviation from target position
2. Excessive control effort
3. Tracking performance

EQUATION 7: LQR Cost Function

$$J = \sum [q_1 \cdot e^2 + q_2 \cdot v^2 + r \cdot u^2]$$

where:

- q_1 = position error weight (how much we penalize position error)
- q_2 = velocity error weight (how much we penalize high velocity)
- r = control effort weight (how much we penalize large servo commands)
- e = position error = $x_{\text{target}} - x_{\text{measured}}$
- v = velocity estimate
- u = servo control signal
- $\sum$ means "sum over all time steps"

The optimal control law that minimizes J is:

EQUATION 8: LQR Control Law

$$u = -K_1 \cdot e - K_2 \cdot v$$

where:

- K_1 = optimal position gain (≈ 0.25 in this project)
- K_2 = optimal velocity gain (≈ 0.15 in this project)
- e = position error

- v = velocity estimate

Interpretation: The servo command is proportional to both the position error AND velocity. If the ball is far from target, we give a strong command. If the ball is moving fast, we dampen that motion.

III.B Kalman Filter Theory

The Kalman filter solves the problem: How do we estimate the true state when we have noisy measurements?

It consists of two steps each cycle:

STEP 1: Prediction

EQUATION 9: Predicted Position

$$\hat{x}_{k|k-1} = \hat{x}_{k-1|k-1} + \Delta t \cdot \hat{v}_{k-1|k-1}$$

EQUATION 10: Predicted Velocity

$$\hat{v}_{k|k-1} = \hat{v}_{k-1|k-1} + \Delta t \cdot K_{k|k-1} \cdot u_{k-1|k-1}$$

EQUATION 11: Prediction Uncertainty (Covariance)

$$P_{k|k-1} = P_{k-1|k-1} + Q$$

where:

- $\hat{x}_{k-}$ = predicted position (before using new measurement)
- $\hat{v}_{k-}$ = predicted velocity
- Δt = time step (0.02 seconds for 50 Hz)
- P_{k-} = uncertainty in prediction
- Q = process noise covariance
- u = control input from previous step

STEP 2: Update (Correction)

EQUATION 12: Kalman Gain

$$K_k = P_{k-} / (P_{k-} + R)$$

where:

- K_k = Kalman gain (how much to trust the new measurement)
- R = measurement noise variance

EQUATION 13: Position Correction

$$\hat{x}_k = \hat{x}_{k-} + K_k \cdot (z_k - \hat{x}_{k-})$$

where:

- z_k = new sensor measurement
- $(z_k - \hat{x}_{k-})$ = innovation (difference between measurement and prediction)

EQUATION 14: Covariance Update

$$P_k = (1 - K_k) \cdot P_{k-1}$$

Interpretation:

- If K_k is close to 1: We trust the sensor (small uncertainty)
- If K_k is close to 0: We trust the prediction (large measurement noise)

III.C Velocity Estimation

Since the Arduino only measures position, velocity must be estimated:

EQUATION 15: Numerical Differentiation (Simple)

$$v_{\text{raw}} = (x_{\text{current}} - x_{\text{previous}}) / \Delta t$$

EQUATION 16: Exponential Smoothing (Applied)

$$v_{\text{estimated}} = \alpha \cdot v_{\text{raw}} + (1 - \alpha) \cdot v_{\text{estimated_previous}}$$

where:

- α = smoothing factor (0.3 in this project)
- Values closer to 1 = more responsive to changes
- Values closer to 0 = more stable filtering

III.D Complete Control Algorithm (Step-by-Step)

Step 1: Read Sensor

- Trigger HC-SR04 ultrasonic sensor
- Measure distance z (in cm)
- Validate range (3.9–12.9 cm)

Step 2: Kalman Filter Update

- Calculate Kalman gain: $K = P / (P + R)$
- Update position: $\hat{x} = \hat{x}_{\text{pred}} + K \cdot (z - \hat{x}_{\text{pred}})$
- Update covariance: $P = (1 - K) \cdot P_{\text{pred}}$

Step 3: Estimate Velocity

- Calculate raw velocity: $v_{\text{raw}} = (\hat{x} - \hat{x}_{\text{previous}}) / \Delta t$
- Apply smoothing: $v = \alpha \cdot v_{\text{raw}} + (1 - \alpha) \cdot v_{\text{previous}}$
- Store current state for next iteration

Step 4: Calculate Error

- $e = x_{\text{target}} - \hat{x}$
- (Usually $x_{\text{target}} = 5$ cm, middle of beam)

Step 5: LQR Control Law

- $$u_{\text{lqr}} = -(K_1 \cdot e + K_2 \cdot v)$$

- u_{lqr} is the "raw" servo command

Step 6: Apply Enhancements

- Nudge boost: If $|u_{lqr}| < 2$, set $u_{lqr} = 2 \cdot \text{sign}(u_{lqr})$ (overcome static friction)
- Saturation: Clamp u_{lqr} to range $[-90, 90]$ (servo limits)
- Rate limiting: Prevent servo from moving too fast
 - $\Delta u_{\max} = 5$ degrees per cycle
 - $u_{\text{new}} = \text{clamp}(u_{lqr}, u_{\text{previous}} - \Delta u_{\max}, u_{\text{previous}} + \Delta u_{\max})$

Step 7: Servo Filtering

- Apply low-pass filter to servo command
- This reduces jitter and vibration

Step 8: Generate PWM

- Convert servo angle (0–180°) to pulse width (1000–2000 microseconds)
- Output to Timer1 PWM

Step 9: Send Telemetry

- Print to UART: position, velocity, error, servo command
- Useful for debugging and performance monitoring

IV. Results from Simulations and Experiments

IV.A Expected Behavior

When the system starts:

1. Initial Phase (0–1 second): Ball position jumps due to sensor startup. Kalman filter rapidly reduces noise.
2. Convergence Phase (1–5 seconds): LQR control actively moves the servo to guide the ball toward the target position. Velocity increases initially, then decreases as the ball approaches target.
3. Steady-State Phase (> 5 seconds): Ball oscillates slightly around the target position (± 0.2 cm). Servo issues small corrective commands. System reaches stable equilibrium.

IV.B Performance Metrics

Metric	Target	Achieved
Settling Time	< 5 seconds	
Steady-State Error	± 0.3 cm	
Noise Reduction	70%	
Rise Time	< 2 seconds	
Overshoot	< 10%	

IV.C Simulation Results

Scenario 1: Ball Starting at $x = 2$ cm

Time (s)	Position (cm)	Velocity (cm/s)	Servo Command (°)
0.0	2.0	0.0	90
0.5	3.2	2.4	85
1.0	4.1	1.8	88
2.0	4.9	0.6	92
3.0	5.0	-0.1	90
5.0	5.0 ± 0.2	0.0 ± 0.05	90 ± 2

Scenario 2: Disturbance (Manual Push)

- At $t = 3$ second, ball is pushed to $x = 7$ cm
- Controller detects error: $e = 5 - 7 = -2$ cm
- LQR generates negative command (servo tilts to push ball back)
- Ball returns to target in ~ 1.5 seconds
- Demonstrates robustness to disturbances

IV.D Experimental Observations

Noise Reduction: Without Kalman filter, position measurements oscillate ± 0.5 cm.

With Kalman filter, oscillation is ± 0.1 cm.

Kalman Gain Steady State:

- Theoretical: $K \approx P/(P + R) \approx 0.5$ (equal trust to prediction and sensor)
- Observed: $K \approx 0.48$ (very close!)

Control Gains Tuning:

- If K_1 too large: Servo oscillates excessively (instability)
- If K_1 too small: Ball converges slowly
- If K_2 too large: Damping is excessive, slow response
- If K_2 too small: System oscillates (underdamped)
- Optimal: $K_1 \approx 0.25$, $K_2 \approx 0.15$ (determined experimentally)

IV.E Limitations and Practical Considerations

Hardware Limitations:

1. Servo has 20 ms response time (50 Hz max frequency)
2. Ultrasonic sensor has ~1–2 cm noise variance
3. Static friction requires minimum servo command ($\sim 2^\circ$)
4. Beam has gravity offset (sensor zero doesn't correspond to horizontal)

Algorithm Limitations:

1. Velocity estimated numerically (noisy, especially at low speeds)
2. Kalman filter assumes linear system (but servo response is nonlinear at small angles)
3. Assumes process and sensor noise are Gaussian (may not be true)
4. Ignores air drag and rolling friction (minor effects)

Embedded System Constraints:

1. ATmega328P has 1 KB static RAM (tight on memory)
2. Limited computational power (~16 MIPS)
3. No floating-point hardware (uses software float library)
4. Serial communication adds latency